

# Airline On-Time Performance

## Preliminary Data Analysis

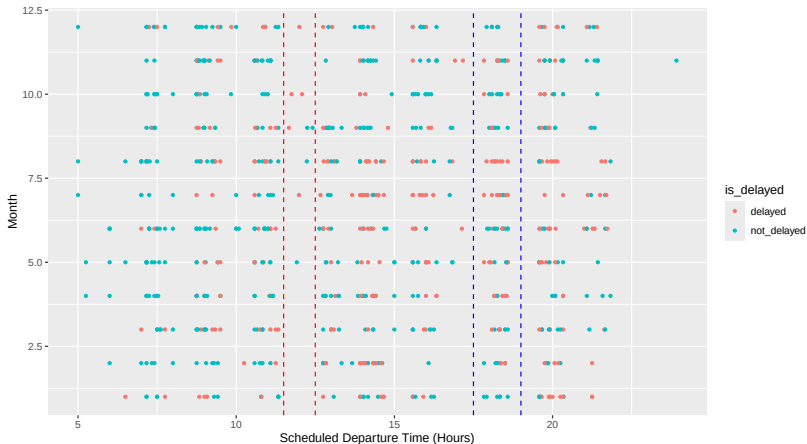
Alex Czerkawski

## Review of scope

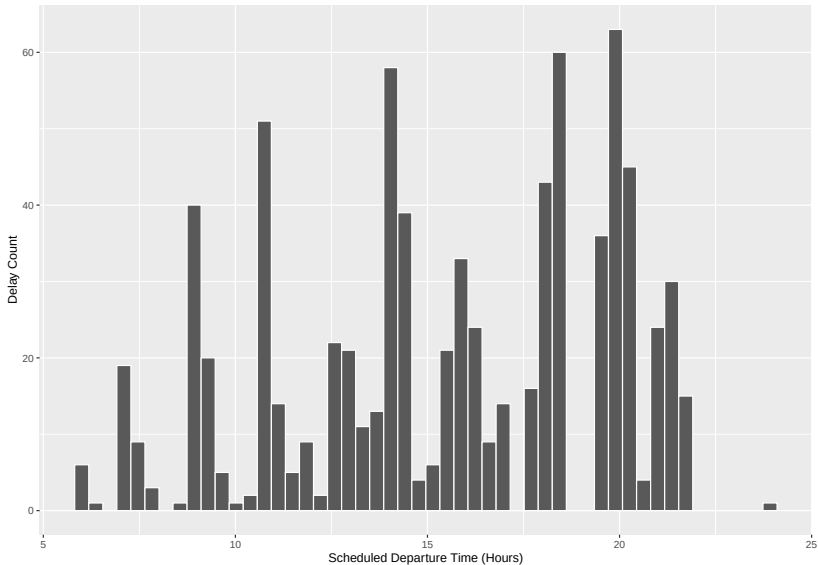
- ▶ Identifying flights at high risk of delay months in advance is a key challenge for United Airlines' network planning team
- ▶ A model will be built using historical flight data to predict the probability of departure delays at ORD (Chicago O'Hare International Airport)
- ▶ Ideally, this model will allow planners to estimate operational impact to make better decisions
- ▶ For the scope of this project, we'll take United Airlines data between October 2024 and September 2025

# Departures by Month and Scheduled Departure Time at ORD

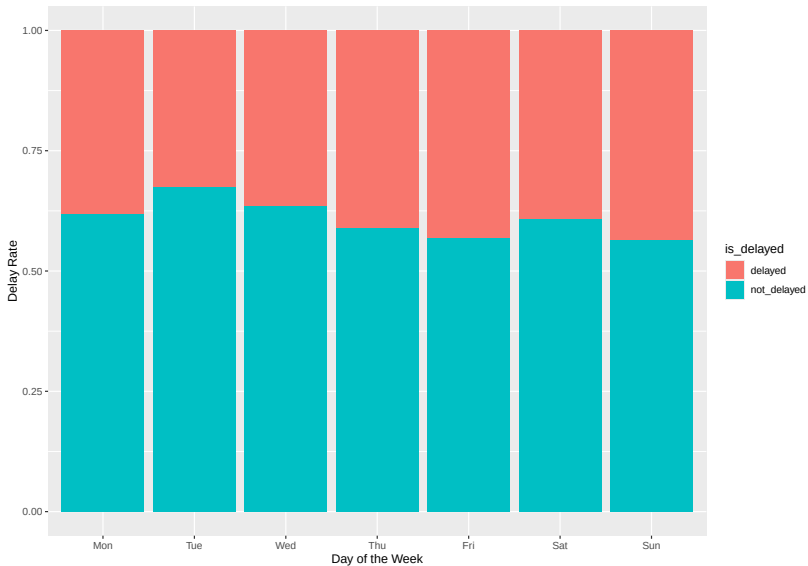
- Random sample of 800 flights was taken



# Delay count by Scheduled Departure Time at ORD



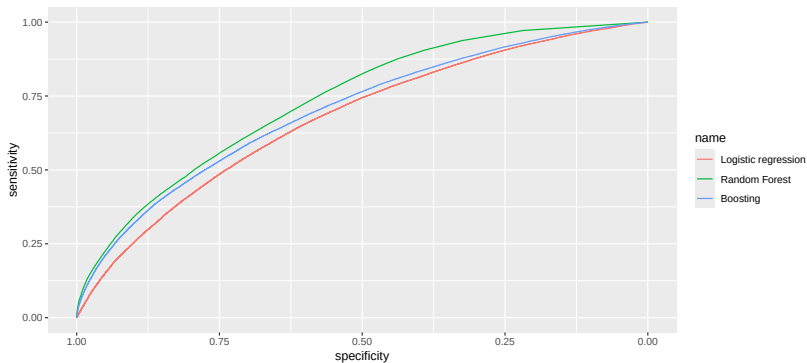
# Delay rate by Day of Week



# Model Fitting

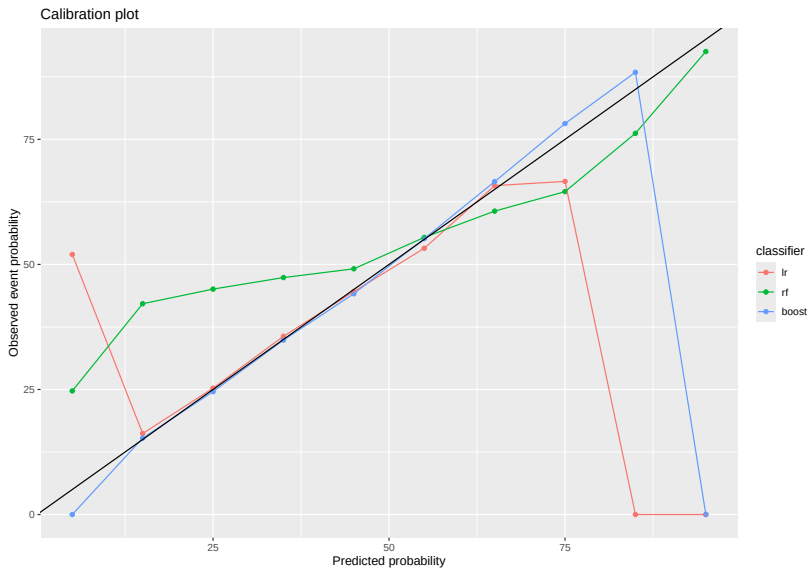
- ▶ % flights on-time = 60.8% (majority class)
- ▶ Predictors include Scheduled Departure Time, Scheduled Arrival Time, Scheduled Block Time, Day of the Week, and Month
- ▶ Three models will be fit, Logistic Regression, Random Forest, and Boosting
- ▶ Models will be evaluated on AUC and calibration
- ▶ Accuracy and kappa aren't as relevant since the goal is predicting an accurate delay rate, not classification

# Model comparison



| Model               | CV_ROC | AUC   | Sensitivity | Specificity |
|---------------------|--------|-------|-------------|-------------|
| Logistic Regression | 0.673  | 0.674 | 0.415       | 0.801       |
| Random Forest       | 0.693  | 0.736 | 0.368       | 0.868       |
| Boosting            | 0.700  | 0.701 | 0.397       | 0.853       |

# Model calibration



## Model calibration continued

| Bin | Logistic Regression |       | Random Forest |       | Boosting |       |
|-----|---------------------|-------|---------------|-------|----------|-------|
|     | Pct                 | N     | Pct           | N     | Pct      | N     |
| 5   | 52.000              | 13    | 24.710        | 22811 | 0.000    | 0     |
| 15  | 16.196              | 1984  | 42.140        | 6635  | 15.260   | 2809  |
| 25  | 25.259              | 11503 | 45.067        | 5271  | 24.576   | 9538  |
| 35  | 35.630              | 13857 | 47.371        | 4127  | 34.866   | 15499 |
| 45  | 44.576              | 12965 | 49.124        | 4205  | 44.138   | 13614 |
| 55  | 53.228              | 17765 | 55.399        | 4689  | 55.155   | 11678 |
| 65  | 65.752              | 10515 | 60.655        | 4850  | 66.554   | 8861  |
| 75  | 66.598              | 323   | 64.555        | 5522  | 78.155   | 5438  |
| 85  | 0.000               | 0     | 76.213        | 8279  | 88.414   | 1488  |
| 95  | 0.000               | 0     | 92.555        | 2536  | 0.000    | 0     |

## Next steps

- ▶ Incorporate more predictors:
  - ▶ Scheduled flights within 30 minutes of departure to capture congestion
  - ▶ Arrival airport to capture market-specific delays
- ▶ Compare to a non-hub airport departure airport like PIT